INCH POUND

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DETAIL SPECIFICATION

CARTRIDGE, DELAY, M90 ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the M90 Delay Cartridge (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4 or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4 or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-A-2550	Ammunition, General Specification for
MIL-A-8625	Coating, Anodic, Conventional for Aluminum and Aluminum Alloy

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-810	Environmental Test Methods and Engineering Guidelines

Comments, suggestions, or questions on this document should be addressed to Commander, Indian Head Division, Naval Surface Warfare Center, Technical Information Branch (Code 4230), 101 Strauss Avenue, Indian Head, Maryland 20640-5035, or emailed to amanda.penn@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil/

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(Copies of these documents are available online at http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications . The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Drawings	
5185050	M90 Closed Bomb Assembly (5.747 in^3)
DL8596123	Cartridge, Delay, M90 Assembly
8845190	Carton, Packing, Ammunition, For Cartridge, Delay: M46, M71, M90, XM102, XM103, XM104, XM105, or M114
8845191	Box, Packing, Ammunition, For Cartridge, Delay: M46, M71, M90, XM102, XM103, XM104, XM105, or M114
9227731	Carton, Alternative, Packing, Ammunition, For Cartridge, Delay: M46, M71, M90, XM102, XM103, XM104, XM105, or M114
851AS110	Ignition Device PVU-1/A
Weapon Specifications	
WS 21535	Ignition Devices, (Percussion), PVU-1/A

(Application for copies should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Attn: Technical Information Branch (Code 4230), Indian Head, MD 20640-5035.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E1742 Standard Practice for Radiographic Examination

(Copies of this document are available online at http://www.astm.org/ or from the American Society for Testing and Materials Customer Service, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

AMERICAN NATIONAL STANDARD INSTITUTE/AMERICAN SOCIETY FOR QUALITY

ANSI/ASQ Zl.4 Sampling Procedures and Tables for Inspection by Attributes

ANSI/ASQ A8402 Quality Management and Quality Assurance - Vocabulary

(Copies of these documents are available online at http://webstore.ansi.org/ansidocstore/default.asp, or from the American Society for Quality, PO Box 3005, 611 E. Wisconsin Ave., Milwaukee, WI 53201-4606.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing

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in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a first article sample shall be subjected to first article inspection (see 4.4, 6.2, and 6.3).

3.2 Conformance to documents. The M90 Delay Cartridge covered by this specification shall be manufactured in accordance with DL8596123 and all documents listed thereon.

3.3 Materials. All materials used in the manufacture of the cartridges shall conform strictly to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.4 Components.

3.4.1 Primary components. For the purpose of this specification, the PVU-1/A Ignition Device, M14 Delay Element, Igniter Charge and the cartridge propellant charge are considered primary components. Only primary components from a single lot shall be used in a production lot of cartridges; however, one primary component production lot may be used in more than one cartridge production lot.

3.4.1.1 Ignition device. The ignition device shall be furnished from lots manufactured within the preceding 24 months or recertified within the preceding 12 months. The ignition device shall be manufactured per Drawing 851AS110 and accepted per WS 21535. Prior to assembly into a cartridge, the ignition device shall be radiographically examined for defects as specified in 4.7.1.

3.4.2 Hardness. The hardness of each component part shall be in accordance with the applicable drawing (see 4.7.2).

3.4.3 Protective finish. The finish of each component part shall be in accordance with MIL-A-8625 and the applicable drawing (see 4.7.3).

3.5 Assembly.

3.5.1 Radiographic examination. Each cartridge assembly shall be radiographically examined for defects (see 4.7.4).

3.5.2 Leak Test. Each assembled cartridge shall exhibit no air leakage greater than 1×10^{-4} standard cubic centimeters per second (see 4.7.5) at a 14.7 psia pressure differential.

3.5.3 Vibration. The assembled cartridge shall withstand vibration as prescribed in MIL-STD-810, Method 514.3, Test Procedure I, Test Condition I-3.2.5 without evidence of any disintegration of the ignition device, propellant, or igniter, or derangement of any component (see 4.7.6 and 6.2).

3.6 Functional requirements.

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3.6.1 Ignition device sensitivity. The ignition device shall fire when drop tested at an energy level of 26 inch-ounces in a sectioned cartridge void of igniter and propellant charge (see 4.7.7.1).

3.6.2 Cartridge performance.

3.6.2.1 Misfires. The cartridge shall not fail to fire.

3.6.2.2 Ballistics. The cartridge, when tested in a closed bomb (Drawing 5185050), shall produce the pressures, delay times (see 6.6.1), and times to P_{max} (see 6.6.2) at temperatures specified in table I (see 4.7.7.2).

TABLE I. *Ballistic requirements.*

Temperature (°F)	Delay time (ms)	Pressure (psig)		Time to P (ms)
		Lower	Upper	
–60 to –70	350 ± 100	2000	2700	12
65 to 75	300 ± 50	2000	2700	12
195 to 205	250 ± 100	2000	2700	12

3.7 Workmanship. Cartridges shall be free of cracks, splits, or other defects which might prevent proper mating with connecting parts or proper functioning of the device.

3.7.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

3.7.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.7.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

4. VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

4.2 Quality assurance terms and definitions. Reference shall be made to ANSI/ASQ A8402 to define the quality assurance terms used.

4.3 Inspections. Inspections shall be in accordance with MIL-A-2550 and ASTM E1742.

4.3.1 Inspection conditions. Unless otherwise specified in the contract or purchase order (see 6.2), inspection conditions shall be as follows:

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a. Temperature: Room ambient 20 to 25 °C (68 to 77 °F)
b. Altitude: Sea level
c. Humidity: 50 percent relative, maximum
d. Atmospheric: 14.7 psia

4.4 First article inspection. At the beginning of regular production, the supplier shall submit a first article sample to a Government approved facility for evaluation. The sample shall consist of 21 assembled cartridges for ballistics testing including 12 for vibration tests, 10 sets of unassembled components for dimensional and visual examination, and a quantity in accordance with 4.4.2, i.e., cartridges assembled without the igniter and propellant charge for ignition device sensitivity test. The first article sample shall be manufactured using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply. When it becomes necessary to make a change in either the ignition device, propellant, or igniter lot after initial production, 12 assembled cartridges from the first production lot thereafter shall be tested in accordance with 4.7.6. The **Government reserves the right to require new first article samples until such time as an acceptable sample is submitted (see 6.3).**

4.4.1 Examination, tests, and facility. After provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or other such facility specified in the contract (see 6.2).

4.4.2 Sample size for ignition device sensitivity. The number of cartridges assembled without igniter or propellant to be submitted for ignition device sensitivity tests shall be in accordance with table II.

TABLE II. *Ignition device sensitivity.*

Lot Size	Sample Size
3 to 15	3
16 to 25	4
26 to 40	5
41 to 65	7
66 to 110	10
111 to 180	15
181 to 300	25
301 to 500	30
501 to 800	35
801 to 1300	40
Over 1300	50

4.4.3 Temperature conditioning for ballistic test. The sample shall be temperature conditioned and tested to determine compliance with 3.6.2.2. The sample for firing shall be conditioned to table III.

TABLE III. *Temperature conditioning.*

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Temperature (°F)	Not vibrated	Vibrated
–60 to –70 65 to 75 195 10 205	3 3 3	4 4 4

4.4.4 First article failure. Failure of the sample to comply with the requirements of the drawings and specifications shall result in first article sample disapproval.

4.5 Quality conformance inspection.

4.5.1 Lot.

4.5.1.1 Submission of product. The product shall be submitted in accordance with ANSI/ASQ Z1.4.

4.5.1.2 Lot size. Unless otherwise specified (see 6.2), the production lot size shall be not more than 600.

4.5.2 Examination and tests. The quality conformance sample shall be subjected to the examinations of 4.6 and all the test procedures of 4.7 except for 4.7.6.

4.6 Examination. All cartridges and components shall be examined visually and radiographically and characteristics defined in accordance with the applicable drawings. One hundred percent examination shall be performed for all critical defects. Examination for major or minor defects shall he performed on a defect basis in accordance with the classification of defects. Four power (4×) magnification may be used in performing visual examination for such defects as burrs and foreign matter. When compliance is in doubt as a result of visual inspection, the characteristic shall be measured or gaged to determine acceptability. All nonconforming items shall be rejected.

4.6.1 Cartridge, Delay, M90 Assembly, Drawing 8596123.

Characteristic	Defect	Method of Inspection
Critical		
1	Any part missing	X-ray/Visual
2	Improper assembly	X-ray/Visual
3	Mixed delay elements or propellant	Visual
4	Metal defective	Visual
5	Crimp missing or incorrect	Visual
6	Marking missing, incorrect, or illegible	Visual
Major		
101	Cartridge profile failure	Gage
102	Diameter of body, max	Gage
103	Diameter of head, max	Gage
104	Location of datum diameter	Gage
105	"0" ring split, cracked or cut	Visual
106	Protective finish inadequate or defective	Visual

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107	Foreign matter present	Visual
108	Burr	Visual
Minor		
201	Total length, max	Gage
202	Case damaged	Visual
203	Varnish inadequate or missing	Visual

4.6.2 Delay Element, M14, Drawing 8596122, covering a detail of 8596123.

Characteristic	Defect	Method of Inspection
Critical		
1	Mixed delay elements	Visual
2	Flash hole missing or obstructed	Visual
3	Ignition device uncrimped or seal missing	Visual
4	Marking missing, incorrect, or illegible	Visual
5	Delay charge not visible through flash hole	Visual
Major		
101	Metal defective	Visual
102	Protective finish inadequate or defective	Visual
103	Burr	Visual
104	Foreign matter present	Visual
Minor		
None defined		

4.6.3 Case, Cartridge, Drawing 8595202, Part No. 8595393 covering a detail of Drawing 8596123.

Characteristic	Defect	Method of Inspection
Critical		
1	Metal defective	Visual
Major		
101	Concentricity of head with body	Gage
102	Straightness of base	Gage
103	Diameter of body, min	Gage
104	Diameter of head, min	Gage
105	Thickness of base flat	Gage
106	Thickness of wall	Gage
107	Location of datum diameter	Gage
108	Protective finish inadequate or defective	Visual
109	Burr or sharp corner	Visual
110	Foreign matter present	Visual

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Minor		
201	Diameter of head, max	Gage
202	Hardness	Gage
203	Total length	Gage
204	Radius or chamfer missing or incorrect	Visual
205	Surface finish improper	Visual

4.6.4 Carton, Packing, Ammunition, Drawing 8845190.

Characteristic	Defect	Method of Inspection
Critical		
None defined		
Major		
None defined		
Minor		
201	Contents loose	Manual/Visual
202	Separator or filler missing	Visual
203	Heat sealer incomplete, loose, or distorted	Visual
204	Cut or damaged through impregnated box	Visual
205	Marking missing, incorrect, or illegible	Visual

4.6.5 Box, Packing, Ammunition, Drawing 8845191.

Characteristic	Defect	Method of Inspection
Critical		
None defined		
Major		
None defined		
Minor		
201	Contents loose	Manual/Visual
202	Strapping missing, loose, broken, or improperly assembled	Visual
203	Contents of box exposed	Visual
204	Box damaged	Visual
205	Marking missing, incorrect, or illegible	Visual
206	Number of cartons incorrect	Visual

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4.6.6 Carton, Alternative, Packing, Drawing 9227731.

Characteristic Defect		Method of Inspection
Critical		
None defined		
Major		
None defined		
Minor		
201	Contents loose	Manual
202	Box broken or damaged	Visual
203	Number of units contained incorrect	Visual
204	Marking missing, incorrect, or illegible	Visual
205	Tape incomplete	Visual

4.7 Test methods and procedures.

4.7.1 Ignition device x-ray. Prior to assembly into the cartridge, the ignition device shall be radiographically examined in accordance with ASTM E1742. Each ignition device shall be x-rayed on boards or trays with the cup side perpendicular to the rays. The x-ray negatives shall be identified by item nomenclature and by lot number. Ignition devices are to be left mounted to the boards or trays until after the x-rays have been reviewed and defects identified to facilitate removal of defective units. After screening, the x-ray negatives with defectives identified, shall be forwarded to the engineering activity designated by the acquiring activity. Interpretation of the radiographic films shall verify the following: missing or abnormally thin charge; charge broken, cracked or porous; cracked or off-center anvil or other misplaced or deformed inert parts; anvil abnormally close to the ignition device cup dome; and all other observable defects in assembly.

4.7.2 Hardness. Ten units from each heat-treated lot of components shall be tested with a hardness tester. Failure of a unit to comply with the requirements of the drawings shall be cause for rejection of the lot of components.

4.7.3 Protective finish. Unless otherwise specified (see 6.2), sampling plans and procedures in the determination of acceptability of coated parts shall be in accordance with the provisions set forth in tables IV and V and MIL-A-8625.

4.7.4 Radiographic examination. All assembled cartridges shall be radiographically examined in accordance with ASTM E1742. The cartridges shall be positioned on their sides for the most revealing exposure. All cartridges shall be identified with serial numbers prior to examination. The serial numbers shall be in consecutive order beginning with the number 001 in each production lot. The cartridges shall be arranged on trays or boards in consecutive numerical order, and each radiograph shall carry a permanent identification of the cartridges displayed thereon. The radiograph plate identification shall include item nomenclature, the complete lot number as stamped on the cartridges, the span of serial numbers displayed, and the contract number. Cartridges having any observable imperfections in assembly shall be cause for rejection. Radiographs of the entire production lot shall accompany the ballistic sample to the activity conducting the production lot acceptance test. Discontinuities in serial numbers shall be

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identified on the x-ray negative. Defective cartridges, found by the vendor, shall be clearly identified on the negative and those defectives removed from the production lot. The serial numbers shall be pen or ink stamped on the cartridge with approximately 1/8-inch-high characters. The serial numbers shall be located apart from the other cartridge markings (see 6.2).

4.7.5 Leak test. Each cartridge shall be subjected to specified pressure differential. The method for determining leakage shall be with a helium mass spectrometer. The helium leak rate shall be converted to standard cubic inches air. Failure to comply with the requirements of 3.3.2 shall be cause for rejection of the cartridge.

4.7.6 Vibration test. Twelve of the 21 assembled cartridges of the first article sample, x-rayed as directed in 4.7.4, properly identified, shall be subjected to vibration. The method and procedures for performing the vibration test shall be in accordance with MIL-STD-810, Method 514.3, Test Procedure I, Test Condition I-3.2.5 (see 6.2). Then the 12 vibrated cartridges shall be x-rayed again. Evidence of any disintegration of the ignition device, propellant, or igniter, or derangement of any component shall be cause for rejection of the sample.

4.7.7 Functional. The following tests shall be performed by the Government.

4.7.7.1 Ignition device sensitivity. A quantity of cartridges, in accordance with table II, void of delay charge igniter and propellant, from the first article sample and each subsequent lot shall be tested for sensitivity. In the event loaded cartridges are selected for test, remove and discard all powder charges from the test cartridge by sectioning the cartridge case around the periphery, 1 inch maximum above the base. Drop test at 13 inches (adjusted) using a steel ball 15/16-inch in diameter, weighing 1.94 ± 0.02 ounces, in a ignition device sensitivity machine with a cartridge holder approved by the acquiring agency. The height of 13 inches shall be adjusted taking into consideration the actual weight of the ball so that the energy level specified in 3.6.1 is achieved. Failure of the sample to comply with the requirements of 3.6.1 shall be cause for rejection of the sample or the lot.

4.7.7.2 Ballistics

4.7.7.2.1 First article. Twenty-one cartridges of the first article sample shall be grouped and temperature conditioned as described in table III and 4.8. Following removal from the conditioning chamber, the cartridge shall be tested as required by 3.6.2.2. Failure of a cartridge to comply with the requirements of 3.6.2.2 shall be cause for sample rejection.

4.7.7.2.2 Production lot acceptance. Twenty-one assembled cartridges from each production lot shall be divided into three equal groups, temperature conditioned, and fired in accordance with 3.6.2.2. If the sample is not evenly divisible by three, the remaining cartridges will be tested at –60 and 205 °F. Failure of a cartridge to comply with 3.6.2.2 shall be cause for rejection of the lot. In addition to the sample of cartridges for the above test, 10 additional cartridges shall be furnished from each production lot for investigative purposes.

4.7.7.3 Retest. There shall be no retests.

4.7.7.3.1 Test failure. If a test failure is attributable to an assignable cause, excluding the test cartridge, the original test results shall be discarded and that test reconducted.

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4.8 Tempe rature conditioning. The conditioning chamber shall be preconditioned to the specified temperature. Once the temperature inside the chamber has stabilized, the samples shall be placed inside and maintained at the specified temperature for 4 hours minimum, 24 hours maximum.

4.8.1 Reconditioning. The sample must be tested within five minutes after removal from the conditioning chamber. If this time is exceeded, the cartridge shall be reconditioned in accordance with 4.8.

4.9 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of section 5.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The assembly is used in cartridge actuated devices for Aircrew Emergency Escape Systems. Since the cartridge was developed specifically for use in military aircraft, there is no commercial application.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and revision letter of this specification
b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced and that MIL-STD-810D shall be used (see 2.1, 3.5.3, and 4.7.6)
c. Whether a first article sample is required (see 3.1, 4.4, and 6.3)
d. Test facility for first article tests (see 4.4.1)
e. Production test activity and lot size (see 4.1 and 4.5.1.2)
f. Any special test conditions (see 4.3)
g. Directions for sampling for classification of defects (see 4.6)
h. Directions for shipping radiographs of ignition devices and of entire lot of cartridge assemblies (see 4.7.1 and 4.7.4)
i. Directions for sampling for protective finish tests (see 4.7.3)
j. Level of preservation and packing (see 5.1 and 6.4)
k. That the safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material," DOD 4145.26M are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a

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Government activity, the safety precaution requirements of 'Ammunition and Explosives Ashore," OP 5 are applicable.

6.3 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors whether the items should be a first article sample or a sample selected from the first production items (see 3.1), and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Contract packaging, packing and marking. The following requirements should be specified in the contract.

6.4.1 Packing and marking. Packing and marking should be Level A or B, as specified by the contracting activity (6.2).

6.4.1.2 Level A. Level A packing and marking should be per Drawings 8845190 and 8845191.

6.4.1.3 Level B. Level B packing and marking should be per Drawings 8845190 and 9227731.

6.5 Hazard notice. The cartridge described herein and certain of its components are flammable or explosive and present hazards in manufacture, handling, storage, and shipment. The contractor should recognize these properties and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling, and electrically induced incidents.

6.6 Definitions.

6.6.1 Delay time. Delay time is defined as the time from primer impact to the start of pressure rise.

6.6.2 Time to P_{max}. Time to P_{max} is defined as the time that elapses between the first indication of pressure to maximum pressure as recorded on a pressure time record.

6.7 Subject term (keyword) listing.

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

6.8 Changes from previous issue. Marginal notations are not used to identify changes with respect to the previous issue because of the extensiveness of the changes. This revision was authorized by ECP number 045220I037.

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